

[2025-03-31] S1P04 – Practical 04.

- **Practical Theme: JavaScript and Bootstrap**
- **Due date: 2025-04-07 before 23:59.**

Activity Notes & Instructions

PURPOSE AND OBJECTIVE.

The purpose and objective of this practical activity is to:

- Objective 1: Creating a ASP.NET MVC (C#) project.
- Objective 2: Creating a model, controller and a view in the project.
- Objective 3: Adding properties and decorators to the model.
- Objective 4: Using the controller and a very simple list data.
- Objective 5: Displaying simple list data in the view.
- Objective 6: Update the global layout menu to include the new view.
- Objective 6: Add new HTML pages with images and style those using Bootstrap.
- Objective 7: Use JavaScript to add interactivity to the View which displays the data.
- Objective 7: Using the GitHub repository to load the complete project.

MATERIAL BEING REFERENCED.

To successfully complete this activity, a student should refer to the notes, details and sample projects that is part of:

- S1L02 Session 02 - Getting started with Visual Studio and GitHub.
- S1L03 Session 03 - Working with the ASP.NET Model-View-Controller.
- S1P02 Practical 02 - Working with the ASP.NET Model-View-Controller.
- S1L04 Session 04 – HTML and CSS within MVC.
- S1L05 Session 05 – Responsive web design using Bootstrap.

ADDITIONAL USEFUL RESOURCES.

Additional reading resources that may assist with the completion of this activity:

- Learn ASP.NET MVC 5 - TutorialsTeacher.com [URL: <https://www.tutorialsteacher.com/mvc>].

RESEARCH THAT WILL BE NECESSARY.

You will have to conduct independent research and readings on the following topics:

- How to add inline event listeners (“onclick”) to HTML elements.
- How to create simple functions in JavaScript
- How to remove an element from a HTML collection
- How to work with parent and child elements
- How to style HTML pages using latest Bootstrap 5.
- How to link to external CSS and/or JS files.

GENERAL INSTRUCTIONS.

Please ensure that you follow the following general instructions.

- This is an individual practical.
- Practical work is based on assessment objectives. If an objective has been achieved a mark will be allocated.
- All practical activities conceptually (based on skills) feed into one or more homework assignments.
- By completing the practical work, you will be learning the necessary skills to complete homework assignments.

SUBMISSION DUE: 2025-04-07 before 23:59.

- There shall be no extensions to the deadline.
- All practical work will be marked in the following practical session during the follow-up week.
- Verify the completeness of your upload.
- Incomplete uploads will be considered unsubmitted work.
- E-mail and / or late submissions WILL NOT be accepted.
- NO EXCEPTIONS WILL BE MADE FOR ANYONE.

ACTIVITY DESCRIPTION AND DETAILS.

For this practical, please ensure that you comply with the instructions provided. The stipulated instructions are mandatory and not optional.

For the following tasks you will have to use:

- ASP.NET Web Application (.NET Framework) for C#, Windows, Cloud and Web.
- ASP.NET Model-View-Controller (C#).

TASKS - MVC:

1. Use your completed Practical 2 project to work from.
2. Restyle two of the HTML Person pages as follows. The other pages can be left as is. On these two pages, provide links to the appropriate Bootstrap and JQuery files (see slides of lecture 5). One of the pages must make use of the Bootstrap grid system. The three sections (Background and bio, hobbies and modules) must now be displayed next to each other. The menu must also be styled using predetermined Bootstrap classes. Any extra styling should be in a separate, external CSS file loaded into the Content folder. (You are allowed to do some of the styling inline).

Person 1 - Name and Surname

[MAIN](#)
[LIST](#)
[Person 01](#)
[Person 02](#)
[Person 03](#)
[Person 04](#)
[Person 05](#)
[BACK](#)

Biography and Background



In this paragraph you have to give the following information about the person in your study group: 1) From where, 2) the degree he/she studies and 3) the reason why he/she studies that degree. For example: Mary Alexander was born in Polokwane, and went to school in Haenertsburg. She studies BIT(IS) with Statistics as other elective because she believes Data Science is the future. Add a photograph of your friend. Make sure that the photo is situated at the left of the text and does not flow into the hobbies section.

Hobbies

In this paragraph you have to give a description of the hobbies of your friend. Insert a photo related to the hobby(ies) of your friend. The photo must be situated at the right of the text.



Modules

In this section you have to provide the first and second semester modules your friend is enrolled for - in two columns.

Modules Semester 1	Modules Semester 2
Module 11	Module 21
Module 12	Module 22
Module xx	Module yy

3. The other page must make use of the predefined Bootstrap “Panel” class and be styled as follows. Any extra styling should be in the external CSS file referred to above.

Person 3 - Name and Surname

[MAIN](#)
[LIST](#)
[Person 01](#)
[Person 02](#)
[Person 03](#)
[Person 04](#)
[Person 05](#)
[BACK](#)

More information on this person

Background and biography



In this paragraph you have to give the following information about the person in your study group: 1) From where, 2) the degree he/she studies and 3) the reason why he/she studies that degree. For example: Mary Alexander was born in Polokwane, and went to school in Haenertsburg. She studies BIT(IS) with Statistics as other elective because she believes Data Science is the future. Add a photograph of your friend. Make sure that the photo is situated at the left of the text and does not flow into the hobbies section.

Hobbies

In this paragraph you have to give a description of the hobbies of your friend. Insert a photo related to the hobby(ies) of your friend. The photo must be situated at the right of the text.



Modules

In this section you have to provide the first and second semester modules your friend is enrolled for - in two columns.

Modules Semester 1

Module 11
Module 12
Module xx

Modules Semester 2

Module 21
Module 22
Module yy

4. Add a column to the View (displaying the list of study buddies) and add a button to the newly created cells.

List of People

Student number	First Name	Last Name	Email address	Link to personal page	Delete
u0001	Name 1	Surname 1	person1@tuks.co.za	LINK	Delete
u0002	Sameer	Surname 2	person2@tuks.co.za	LINK	Delete
u0003	Name 3	Surname 3	person3@tuks.co.za	LINK	Delete
u0004	Name 4	Surname 4	person4@tuks.co.za	LINK	Delete
u0005	Name 5	Surname 5	person5@tuks.co.za	LINK	Delete

5. By clicking on a button, the corresponding row is removed. Hint: you need to find more info on, the “onclick” event listener, how to remove an element from a HTML collection, how to work with parent and child nodes.
6. Use JavaScript to change the colour of the text to red in the row with the info of your study buddies whose first names begin with a “S” – or any other letter you prefer.

List of People

Student number	First Name	Last Name	Email address	Link to personal page	Delete
u0001	Name 1	Surname 1	person1@tuks.co.za	LINK	Delete
u0002	Sameer	Surname 2	person2@tuks.co.za	LINK	Delete
u0003	Name 3	Surname 3	person3@tuks.co.za	LINK	Delete
u0004	Name 4	Surname 4	person4@tuks.co.za	LINK	Delete
u0005	Name 5	Surname 5	person5@tuks.co.za	LINK	Delete

03. [2025-03-31] S1P03 – Practical 03.

- **Practical Theme:** JavaScript and Responsive Web design with Bootstrap
- **Due date:** 2025-04-07 before 23:59.

Rubric [____ / 20]

Student Number									Surname	Initials

REQUIREMENT [Marking].	DYSFUNCTION	PARTIAL	MAX
• Check the appropriate mark [XX]. • Add final mark and assign to top of rubric.			
1. New project with appropriate project name.	0	1	2
2. One HTML page with correct layout – three columns	0	1	2
3. One HTML page with correct layout - panes	0	1	2
4. Menu restyled using Bootstrap on both pages	0	0	1
5. Other styling done and stored in external CSS file in Content folder	0	1	3
6. On View page – added new column with buttons - styled correctly	0	1	2
7. Create function to delete the appropriate row	0	2	4
8. Use JavaScript to change text colour to red of appropriate rows.	0	2	4
TOTAL	0	9	20